

JIEUN SUNG

Ph.D. Candidate in Computational Biology and AI for Drug Discovery

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EDUCATION

Ewha Womans University

Sep 2021-Feb 2027 (Expected)

Ph.D. Candidate in Life Science, Specialization in Bioinformatics

- Systems Pharmacology Laboratory, Department of Life Science

Ewha Womans University

Mar 2017-Aug 2021

B.S. in Chemistry with a Double Major in Bioinformatics

Honors: Graduated with Honors (GPA 3.84/4.3); Academic Excellence Scholarship

RESEARCH INTERESTS

AI-driven drug discovery; single-cell perturbation modeling and transcriptomic response prediction; drug-target interaction prediction; cellular reprogramming and gene regulatory networks; biological foundation models.

PUBLICATIONS

1. **Sung, J.** and Kim, W. (2026). scDEBART: Predicting *in silico* Single-Cell Perturbation Responses via Large-Scale Differential Expression Learning. In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*. [OpenReview](#).
2. **Sung, J.**, Kim, S., and Kim, W. TDEP: Predicting Target Perturbation Response from Drug-Induced Transcriptomes. Manuscript under review.
3. **Sung, J.***, Son, K.*, Cha, H.-J., Kim, W., and Lee, H. FateDriver: A Dual-View Graph Neural Network for Predicting Driver Genes in Cellular Reprogramming. Manuscript in preparation.
4. **Sung, J.** and Kim, W. TDEM: Deconvolving Drug-Induced Transcriptomes into Target-Specific Perturbation Profiles via Matrix Factorization. Manuscript in preparation.

* Equal contribution.

PATENTS

1. **Sung, J.** and Kim, W. Large-Scale Pretrained Transformer Model and Method for Predicting Expression Changes (logFC) from Single-Cell Transcriptomes. Republic of Korea provisional patent application No. 10-2026-0094556, filed May 26, 2026. Assignee: Ewha Womans University.

HONORS & AWARDS

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| Jul-Aug 2025 | Third Place, 3rd AI Drug Discovery Competition
Placed 3rd among 1,137 participants in a national competition focused on predicting MAP3K5 inhibitor IC50 values from molecular structures. Hosted by the Korea Pharmaceutical and Bio-Pharma Association and sponsored by the Ministry of Health and Welfare, Yuhan Corporation, and CAS. |
| Aug-Sep 2024 | Second Place, 2nd AI Drug Discovery Competition
Placed 2nd among 1,798 participants in a national competition focused on predicting IRAK4 inhibitor IC50 values from molecular structures. Hosted by the Korea Pharmaceutical and Bio-Pharma Association's AI Drug Convergence Research Institute and sponsored by the Ministry of Health and Welfare, Korea Health Industry Development Institute, and Daewoong Pharmaceutical. |

RESEARCH FUNDING

National Research Foundation of Korea (NRF) Ph.D. Student Research Promotion Grant

AI-Based Modeling of Drug-Induced Transcriptomic Changes

Funding period: Sep 2024–Aug 2026

Principal Investigator | Total funding: KRW 50,000,000.

PRESENTATIONS

Invited Talks

1. **Sung, J.** Building AI Prediction Models from Biological Data. Invited lecture, Systems Biology Lab, Department of Pharmacy, Pusan National University, Busan, Republic of Korea, February 2026.

Contributed Talks and Posters

1. **Sung, J.**, Son, K., Cha, H.-J., Kim, W., and Lee, H. FateDriver: A Dual-View Graph Neural Network for Predicting Driver Genes in TF-Induced Cellular Reprogramming. Annual Meeting of the Korean Society for Stem Cell Research (KSSCR), Republic of Korea, August 2026. Poster.
2. **Sung, J.** and Kim, W. scDEBART: Predicting *in silico* Single-Cell Perturbation Responses via Large-Scale Differential Expression Learning. 43rd International Conference on Machine Learning (ICML 2026), COEX Convention & Exhibition Center, Seoul, Republic of Korea, July 2026. Poster.
3. **Sung, J.**, Son, K., Cha, H.-J., Kim, W., and Lee, H. FateDriver: A Dual-View Graph Neural Network for Predicting Driver Genes in TF-Induced Cellular Reprogramming. Annual Meeting of the Korean Society for Aging Research, Republic of Korea, June 2026. Poster.
4. **Sung, J.**, Kim, S., and Kim, W. TDEP: Target-Perturbed Differential Expression Profiles for Interpretable Drug–Target Interaction Prediction. Annual Meeting of the Korean Society for Bioinformatics (KSBI), Republic of Korea, October 2025. Poster.
5. **Sung, J.** and Kim, W. Uncertainty-Aware BLAST-Weighted Multi-Representation Ensemble Model for IC50 Prediction. 3rd AI Drug Discovery Competition, Republic of Korea, September 2025. Poster.
6. **Sung, J.**, Kim, S., and Kim, W. AI-Based Deconvolution of Drug-Induced Transcriptomic Changes into Single-Target Effects across Multiple Cell Lines. EJO Symposium, Republic of Korea, December 2024. Poster and oral presentation.
7. **Sung, J.** and Kim, W. Ensemble Learning–Based Prediction of IRAK4 Inhibitory Activity (IC50). 2nd AI Drug Discovery Competition, Republic of Korea, October 2024. Poster.
8. **Sung, J.** and Kim, W. Deconvolution of Transcriptomic Changes Caused by Drug Perturbation Using Matrix Factorization. Annual Meeting of the Korea Genome Organization (KOGO), Republic of Korea, January–February 2024. Poster.
9. **Sung, J.** and Kim, W. Deconvolution of Transcriptomic Changes Caused by Drug Perturbation Using Matrix Factorization. Annual Meeting of the Korean Society for Bioinformatics (KSBI), Republic of Korea, November 2023. Poster.
10. **Sung, J.** and Kim, W. Deconvolution of Transcriptomic Changes Caused by Drug Multi-Target Perturbation Using Matrix Factorization. Annual Meeting of the Korean Society for Biochemistry and Molecular Biology (KSBMB), Republic of Korea, May 2023. Poster.
11. **Sung, J.** and Kim, W. Deconvolution of Transcriptomic Changes Caused by Drug Multi-Target Perturbation Using a Neural Network. Annual Meeting of the Korean Society for Bioinformatics (KSBI), Republic of Korea, October 2022. Poster.

TECHNICAL EXPERTISE

Programming: Python, R, Bash, Git, LaTeX

Machine Learning & AI: PyTorch, TensorFlow, Keras, Hugging Face Transformers, DeepSpeed; Transformer architectures, graph neural networks, autoencoders, multi-GPU distributed training

Computational Biology: Single-cell RNA-seq analysis (Scanpy, AnnData, scvi-tools), Bulk RNA-seq analysis (FastQC, fastp, STAR, Salmon, featureCounts, HTSeq), perturbation transcriptomics (Perturb-seq; PerturbScore), differential expression analysis (DESeq2, edgeR, limma), network analysis (STRING, NicheNet), pathway enrichment (clusterProfiler, fgsea), Visualization (ComplexHeatmap, ggplot2)

Computational Drug Discovery: Molecular property prediction, drug–target interaction prediction, molecular docking, virtual screening, protein structure prediction (AlphaFold)